

Lecture 1

§1 Introduction

Let $\mathbb{F}_q$ = finite field with q elements.

Assume $q \geq 5$ odd.

Today we take $G = GL_2(\mathbb{F}_q)$

Q: Can we find the rep theory (at least for now)
character table for G ?

First $|G| = (q^2 - 1)(q^2 - q) = q(q-1)^2(q+1)$

The conjugacy classes are given by rational canonical form.

1. $\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$ with $a \in \mathbb{F}_q^\times \rightarrow q-1$ classes, each of size 1
2. $\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$ with $a \in \mathbb{F}_q^\times \rightarrow q-1$ classes, each of size $(q-1)(q+1)$
3. $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ with $a, b \in \mathbb{F}_q^\times$,
 $a \neq b \rightarrow \frac{(q-1)(q-2)}{2}$ classes, each of size $q(q+1)$
stabilizer is $\begin{pmatrix} a & b \\ 0 & a \end{pmatrix} \rightarrow q(q-1)$
4. $\begin{pmatrix} x & \varepsilon y \\ y & x \end{pmatrix}$ $x \in \mathbb{F}_q, y \in \mathbb{F}_q^\times, \varepsilon \in \mathbb{F}_q^\times \setminus (\mathbb{F}_q^\times)^2$ fixed

$\Rightarrow \frac{q(q-1)}{2}$ classes of size $q(q-1)$

So we have a total of

$$q-1 + q-1 + \frac{(q-1)(q-2)}{2} + \frac{q(q-1)}{2} \\ = q^2 - 1 \text{ classes}$$

$\Rightarrow q^2 - 1$ irreps of G .

What are they? Commutator subgroup of G is $SL_2(\mathbb{F}_q)$

$\Rightarrow \exists q-1$ 1-dim reps of G by $\chi \circ \det$ for $\chi: \mathbb{F}_q^\times \rightarrow \mathbb{C}^\times$

Now let

$$B = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G \right\} \supset \mathcal{U} := \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \right\}$$

Given $\chi_1, \chi_2: \mathbb{F}_q^\times \rightarrow \mathbb{C}^\times$ homs, define a 1-dim rep $\mathbb{C}_{\chi_1 \chi_2}$ of B by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} v = \chi_1(a) \chi_2(d) v, \quad v \in \mathbb{C}$$

Then define

$$\begin{aligned} V_{x_1, x_2} &= \text{Ind}_B^G \mathbb{C}_{x_1, x_2} \\ &= \mathbb{C}[G] \otimes_{\mathbb{C}[B]} \mathbb{C}_{x_1, x_2} \quad q+1\text{-dim} \end{aligned}$$

Can prove

- V_{x_1, x_2} irreducible unless $x_1 = x_2$ in which case $V_{x, x} = x \cdot \det \oplus \text{Sp}_x$ with Sp_x a q -dim rep.
- $V_{x_1, x_2} \cong V_{x'_1, x'_2} \Leftrightarrow \{x_1, x_2\} = \{x'_1, x'_2\}$

So we have

- $q-1$ q -dim irreps Sp_x , called special
- $\frac{(q-1)(q-2)}{2}$ $q+1$ -dim irreps V_{x_1, x_2} called principal series

What are the other $\frac{q(q-1)}{2}$ irreps?

How are they related to

$$S := \begin{pmatrix} x & \epsilon y \\ x & x \end{pmatrix} \cong \mathbb{F}_{q^2}^\times \subseteq G?$$

It turns out that for every character

$$\Theta \in S \cong \mathbb{F}_{q^2}^\times \rightarrow \mathbb{C}^\times$$

st. Θ does not factor through

$$N_m \cong \mathbb{F}_{q^2} \rightarrow \mathbb{F}_q$$

$$x \mapsto x \cdot c(x) \quad \text{with } 1 \notin c \in \text{Gal}(\mathbb{F}_{q^2}/\mathbb{F}_q)$$

there is an ^{assoc} irrep V_Θ of dim $q-1$

$$\text{with } V_\Theta \cong V_{\Theta'} \Leftrightarrow \Theta' = \Theta \circ c$$

$$\Rightarrow \frac{(q^2-1) - (q-1)}{2} = \frac{q(q-1)}{2} \quad \text{called cuspidal}$$

What is V_Θ ?

One description: Fix Θ . Fix also

a nontrivial $\chi: \mathbb{U} \rightarrow \mathbb{C}^\times$. $Z = \left\{ \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} : a \in \mathbb{F}_q^\times \right\}$

Then character of V_Θ is the character of the virtual rep

$$\text{Ind}_{Z\mathbb{U}}^G \Theta|_Z \boxtimes \chi - \text{Ind}_T^G \Theta$$

$\dim = q^2 - 1$
 $\dim = \frac{q(q-1)(q+1)}{(q-1)(q+1)} = q(q-1)$

Huh??

Drinfeld, Deligne-Lusztig: This rep appears naturally in cohomology. Moreover, gives you natural integral lattices in such reps.

A concrete application for number theorists: Congruences between modular forms.

Say $f \in S_2(\Gamma_0(p^2))$ a newform whose associated ℓ -adic, $\ell \neq p$, Gal rep is unramified at p .
Eichler-Shimura + Local Langlands

$\Rightarrow$ Hecke eigensystem of f appears in

$$H^1(\mathrm{SL}_2(\mathbb{Z}), V_\Theta)$$

for some $\Theta: S \rightarrow \mathbb{C}^\times \cong \overline{\mathbb{Q}}_\ell^\times$

Now say we have a mod ℓ Hecke eigensystem $\lambda: \Pi \rightarrow \overline{\mathbb{F}}_\ell^\times$, $m = \dim(\lambda)$

and we want such an f lifting it.
 If $L_0 \subset V_0$ is a G -stable $\mathbb{Z}_\ell$ -lattice,
 then it would be enough to show

$$H^1(\mathrm{SL}_2(\mathbb{Z}), L_0)_m \rightarrow H^1(\mathrm{SL}_2(\mathbb{Z}), L_0 \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell)_m$$

is surj, or that

$$H^1(\mathrm{SL}_2(\mathbb{Z}), L_0)_m = 0$$

Upshot Understanding a lattice L_0 is the DL-rep
 $V_0 \Rightarrow$ certain lifts of mod ℓ Hecke eigenvalues to
 char 0.

This course: Representation theory of
 $\mathrm{SL}_2(\mathbb{F}_q)$ and an excuse to learn some
 of the theories of

- fin groups of Lie type
- étale cohomology
- Deligne-Lusztig theory
- modular rep theory

through a particular example.